Improve and Challenge

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

Recall and choose your starting point

> Record your starting explanation for Improve and Challenge before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Test Diverse Users

Without opening the old lesson, explain one key idea from Test Diverse Users. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Project Problem and Success Test

Without opening the old lesson, explain one key idea from Project Problem and Success Test. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Evidence and Sources

Without opening the old lesson, explain one key idea from **Evidence** and Sources. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 26

Practise with fading help

> Complete the supported Improve and Challenge trace and record the first step you can explain without help.

Use W2: Test neighbors for Improving without hiding failures. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 26

Individual reasoning check

> Check the proposed Improve and Challenge explanation against the supplied facts before discussing it.

Use the worked case for Improving without hiding failures. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 26

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Repair independently

Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 26

Your learning target

> Connect the Improve and Challenge target to a trace and a justified conclusion.

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

WORKBOOK · 6 OF 26

Solve the given case

> Solve the supplied Improve and Challenge case and show the deciding evidence.

A repair raises success from 7/10 to 9/10 on the same validation set.

What belongs in the improvement claim?

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 26

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

After changing a scoring weight, the system fixes one request but breaks two previously correct ones. What should the improvement record say?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 26

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 26

Project checkpoint

> Justify what Improve and Challenge adds to Transparent Book Recommender and keep the evidence.

Publish the cause, one change, fixed test set, **before/after evidence**, and remaining limitation.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 26

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 26

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

After a scoring change, four old failures pass but two previous passes fail. On the original ten cases there were six passes. Compute the new total and explain what else should be reported.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 26

W1 Locate the cause

> Locate the cause. Apply the supplied Improve and Challenge facts and keep your reasoning.

Why does minutes < limit reject a valid exactly-at-limit activity?

My reasoning and evidence

WORKBOOK · 13 OF 26

W2 Test neighbors

> Test neighbors. Apply the supplied Improve and Challenge facts and keep your reasoning.

For limit 15, give expected eligibility at 14, 15 and 16.

My reasoning and evidence

W3 Reject a bad fix

> Reject a bad fix. Apply the supplied Improve and Challenge facts and keep your reasoning.

A proposed fix accepts minutes <= limit + 1. Give a counterexample at limit 15.

My reasoning and evidence

W4 Repair independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Repair independently. Apply the supplied Improve and Challenge facts and keep your reasoning.

Write the eligibility predicate for at most **12 minutes** and a **regression** set including 11,12,13 and 0. Add one chooser tie case and one invalid-data case to rerun.

My reasoning and evidence

W5 Write the change log

> Write the change log. Apply the supplied Improve and Challenge facts and keep your reasoning.

Record the bug, change, **evidence** and one remaining limit without claiming all bugs are gone.

My reasoning and evidence

Testing functions and interfaces — W1 Choose test scale

> Use the supplied Improve and Challenge task to record your reasoning, evidence and next step.

Classify testing score(3,2) versus running raw rows through the whole chooser.

My reasoning and evidence

Testing functions and interfaces — W2 Calculate an oracle

> Use the supplied Improve and Challenge task to record your reasoning, evidence and next step.

What expected score should a test use for fun 4 and learning 3?

My reasoning and evidence

WORKBOOK · 19 OF 26

Testing functions and interfaces — W3 Repair a weak test

> Use the supplied Improve and Challenge task to record your reasoning, evidence and next step.

A test compares recommend(items,10) to itself. Replace it with a useful assertion.

My reasoning and evidence

WORKBOOK · 20 OF 26

Testing functions and interfaces — W4 Test independently

> Use the supplied Improve and Challenge task to record your reasoning, evidence and next step.

Design an integration **fixture**: A valid (minutes 10, fun 2, learning 4), B valid (10,3,2), C invalid (5,9,1), limit 10. State winner, counts and a reversed-order expectation.

My reasoning and evidence

WORKBOOK · 21 OF 26

Testing functions and interfaces — W5 Cover absence

> Use the supplied Improve and Challenge task to record your reasoning, evidence and next step.

> What expected result should an empty-**input** integration case specify?

My reasoning and evidence

WORKBOOK · 22 OF 26

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 23 OF 26

Pictorial case practice: Improve and Challenge

> A fictional recommender excludes books at a page limit that is intended to include equality. Explain the labelled step before checking the changed condition.

A fictional recommender excludes books at a page limit that is intended to include equality.

Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

AI ACADEMY / AI-2/35

Improve and Challenge

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Identify cause

A wrongly excluded at equality, so old code selects B.

2. Focused repair

Replace <100 with <=100 without changing ranking or type checks.

3. Expected revised result

A becomes eligible and wins; C remains excluded.

4. Challenge repair

Rerun 99,100,101; invalid pages; tied scores; reversed row order.

CHANGE ONE THING

An alternative patch simply increases the limit to 101.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why are tests at both 100 and 101 needed?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Transparent Book Recommender record.

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	

Evidence to keep	My record
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 25 OF 26

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender excludes books at a page limit that is intended to include equality.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Rerun 99,100,101; invalid pages; tied scores; reversed row order.

1. Identify cause

2. Expected revised result

3. Challenge repair

4. Focused repair

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Challenge repair Rerun 99,100,101; invalid pages; tied scores; reversed row order.

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: An alternative patch simply increases the limit to 101. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. C is wrongly admitted as well as A.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: C is wrongly admitted as well

as A. Changing the maximum alters the contract instead of repairing its comparison.

Repair a missing building block

Review Test Diverse Users (ai-2/34). Goal: Evaluate across varied cases. Short practice: Can these records establish performance for different user groups?. Return to this week and explain what you changed.

Review Project Problem and Success Test (ai-2/32). Goal: Define a small application purpose. Short practice: Write an equality test and explain which requirement it checks.. Return to this week and explain what you changed.

Review **Evidence** and Sources (ai-2/29). Goal: Explain the **mechanism** and terminology of claim support and citation evaluation.

Short practice: Why is the original answer not fully supported despite having citations on every claim?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

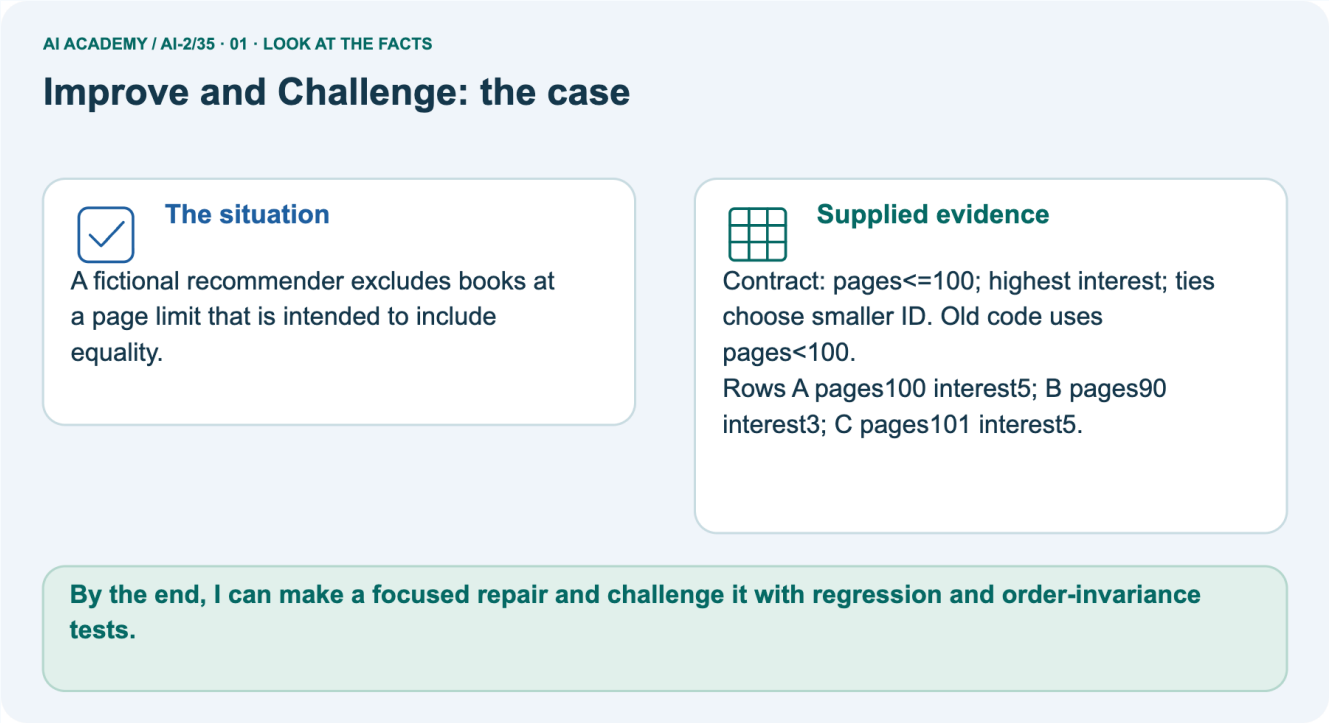
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Improve and Challenge picture from its labels, then record a relevant limit.

01. Improve and Challenge: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

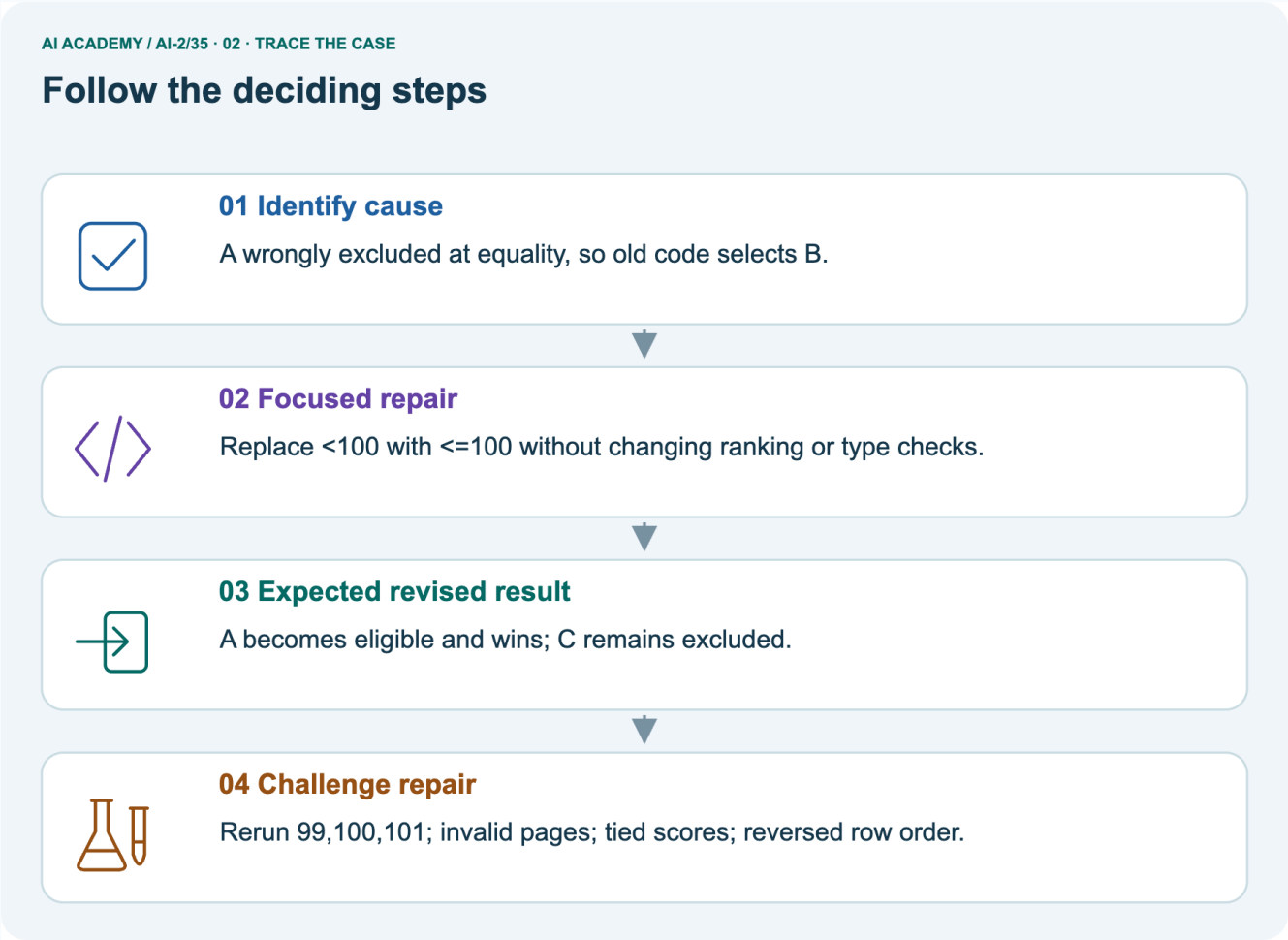
A fictional recommender excludes books at a page limit that is intended to include equality. Supplied facts: Contract: pages<=100; highest interest; ties choose smaller ID. Old code uses pages<100.

Rows A pages100 interest5; B pages90 interest3; C pages101 interest5.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



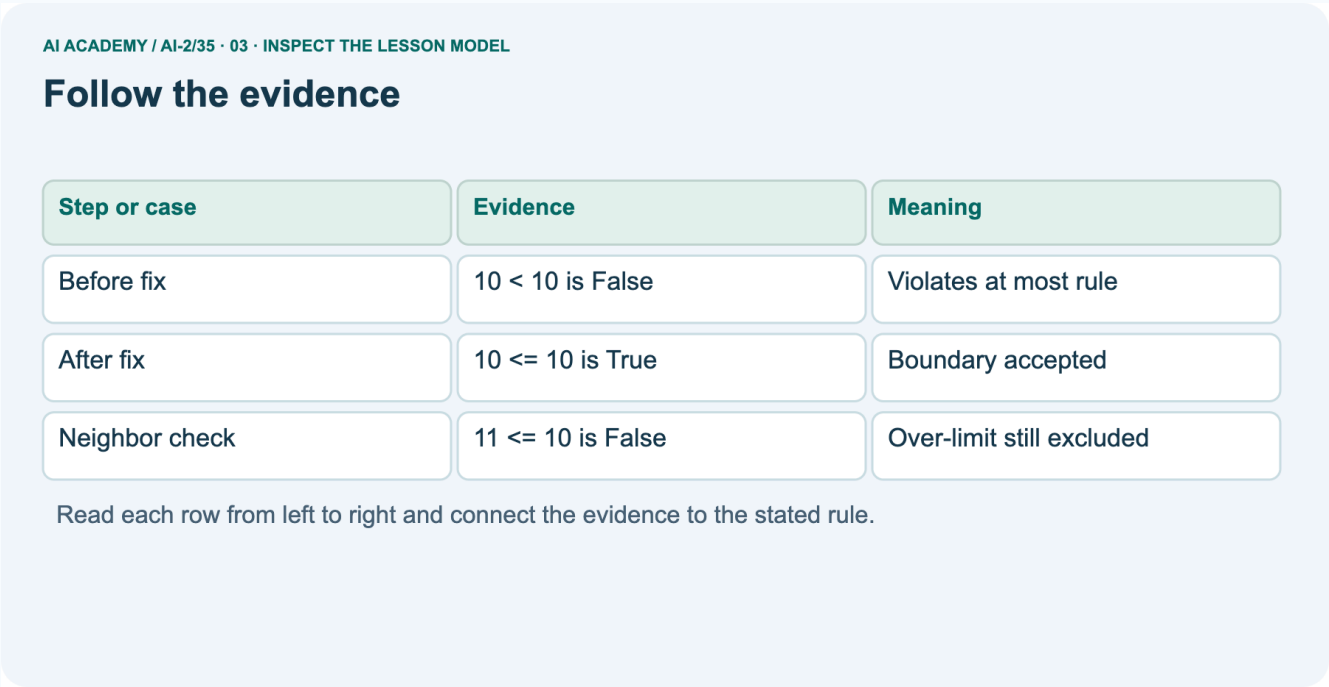
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify cause: A wrongly excluded at equality, so old code selects B. → Focused repair: Replace <100 with <=100 without changing ranking or type checks. → Expected revised result: A becomes eligible and wins; C remains excluded. → Challenge repair: Rerun 99,100,101; invalid pages; tied scores; reversed row order.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence



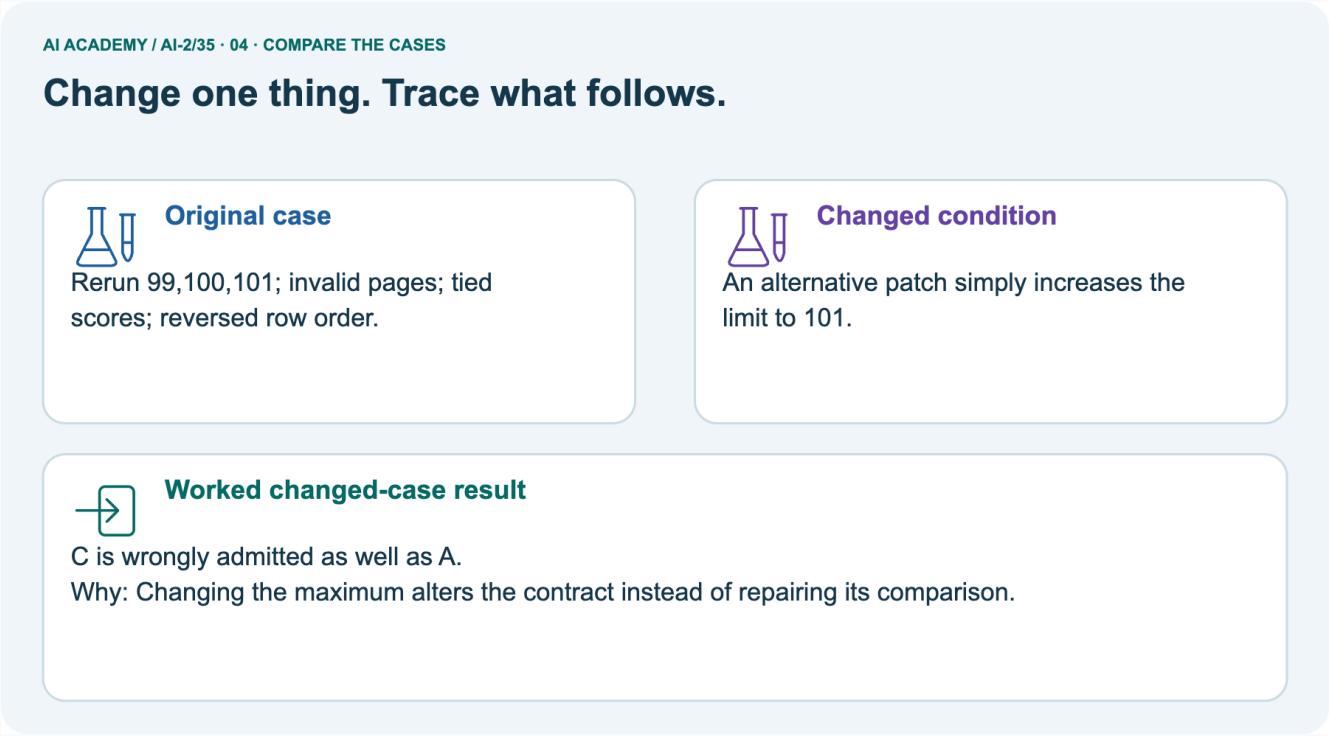
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Before fix | 10 < 10 is False | Violates at most rule. After fix | 10 <= 10 is True | Boundary accepted. Neighbor check | 11 <= 10 is False | Over-limit still excluded

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

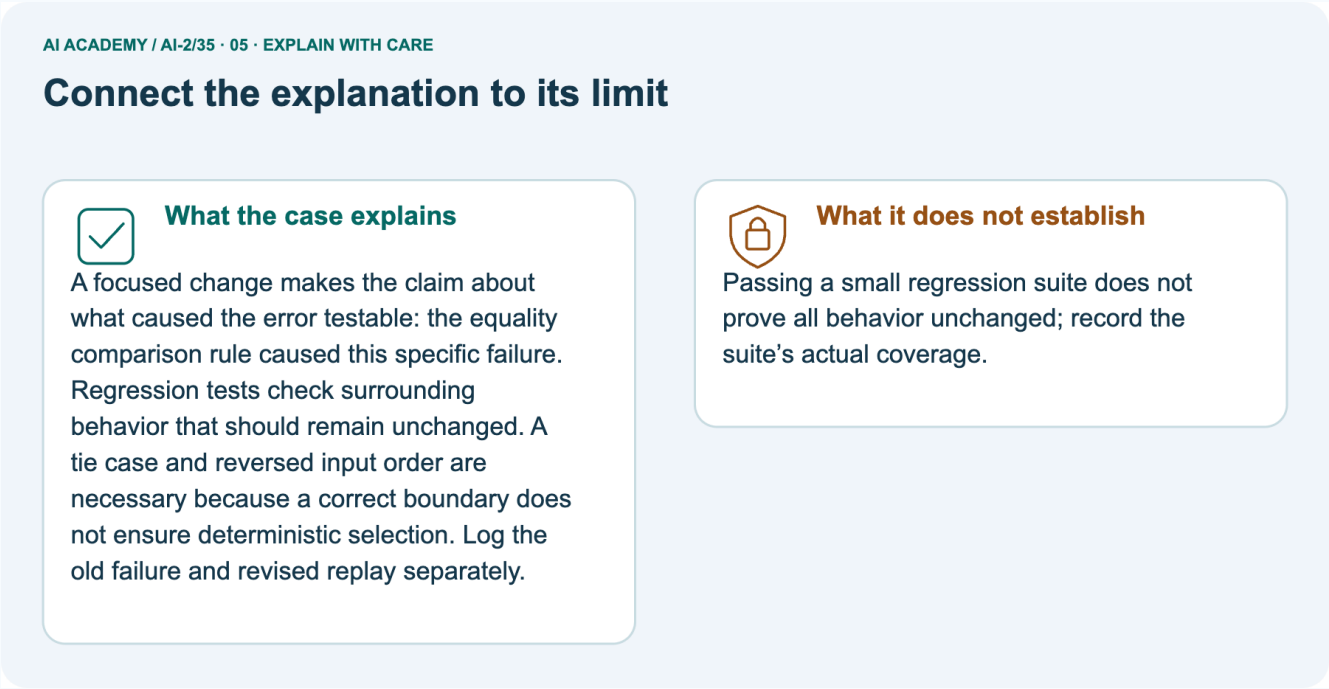
Original case: Rerun 99,100,101; invalid pages; tied scores; reversed row order. Changed condition: An alternative patch simply increases the limit to 101. Expected result: C is wrongly admitted as well as A.

Reason: Changing the maximum alters the contract instead of repairing its comparison.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A focused change makes the claim about what caused the error testable: the equality comparison rule caused this specific failure. **Regression** tests check surrounding behavior that should remain unchanged.

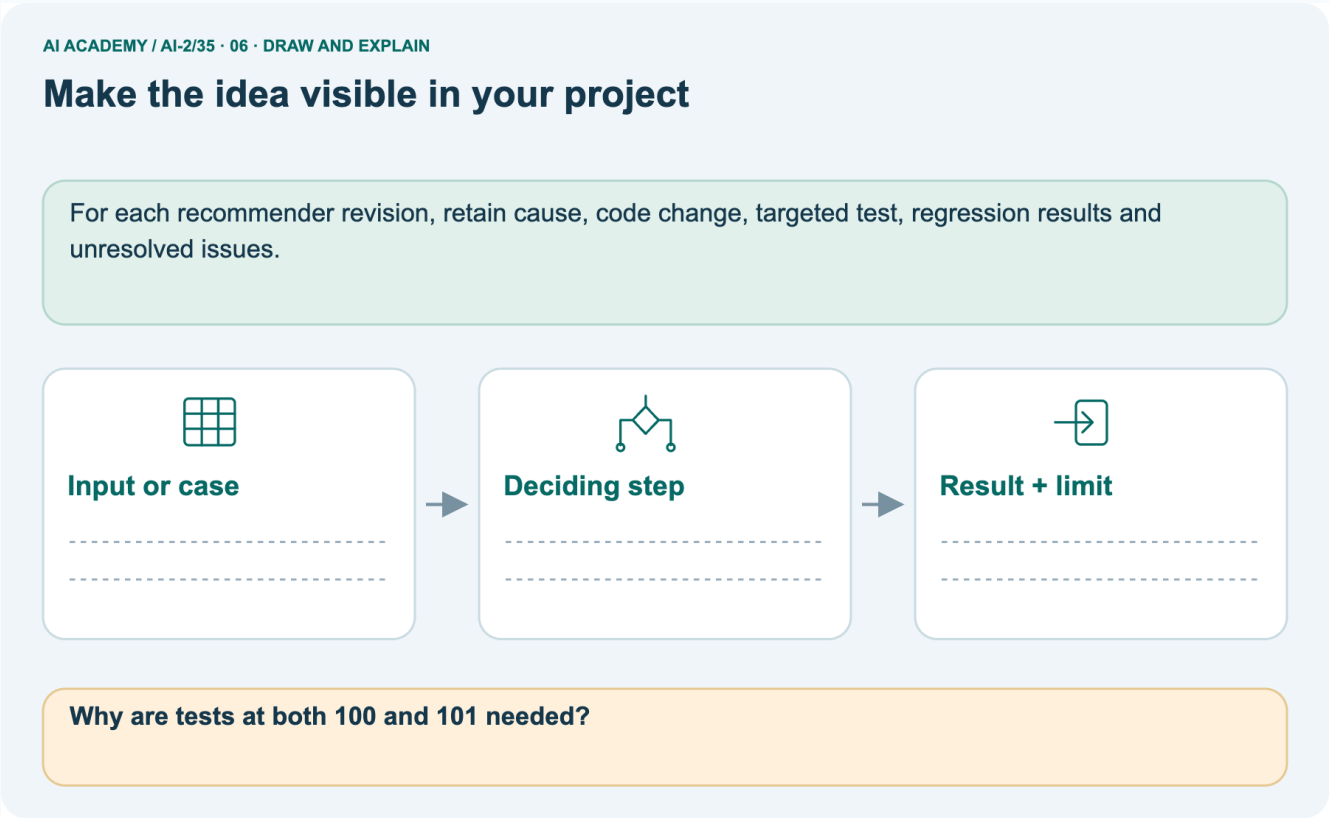
A tie case and reversed **input** order are necessary because a correct boundary does not ensure deterministic selection. Log the old failure and revised replay separately.

Limit: Passing a small **regression suite** does not prove all behavior unchanged; record the suite's actual coverage.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For each recommender revision, retain cause, code change, targeted test, **regression** results and unresolved issues. Draw your **input**, deciding step, result and limit. Supported

question: Why are tests at both 100 and 101 needed?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.